EPSR-test-ex111 Three-body potentials: several groups, several angles

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises several simultaneous three-body groups in the `epsr` engine [1,2,3]: inter- and intramolecular triplets, more than one requested angle per group, negative amplitudes and lines that must be ignored.

Code exercised

Fortran (original, fortran_src/):

- `threebody_mod.f90`: `find_triplets`, `en_mole_triplets`, `pr_mole_triplets`, `save_threeb_energies`, `get_triplet_pot_cons`
- `read_and_save_files.f`: `read_pcof` (`tb` lines), `save_files_others` (re-reads and writes them)
- `ato_manipulation_rev4.f`: `fmolec1`, `press` (three-body energy/pressure); `molfit.f`: `en_change`

Rust (port, rust/epsr-core/src/):

- `threebody.rs`: `read_pcof_threebody`, `write_pcof_threebody`, `find_triplets`, `en_mole_triplets`, `pr_mole_triplets`, `save_threeb_energies`, `en_change_triplets`
- `energy.rs`: `fmolec1`, `en_change`, `press`; `pcof.rs`: `read_pcof`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. A `tb A1 A2 A3 amp awidth rwidth rmax12 rmax32 s12 s23 s13 ang1 amp1 ...` line in the `.pcof` adds an angular energy `amp*ampN*0.5*((cos theta - cosN)/awidth)²` to every triplet A1-A2-A3 (A2 the centre) within the distance limits, for the requested angle N closest to theta; the flags select same or different molecules. Groups: water O1···O1···O1network angle on three different molecules (109.5° and 60°); intramolecular water H1-O1-H1 (104.5°); water O1-H1···O(DMSO)with five angles (180°-100°, one amplitude negative).

Inputs

EPSR-test with five lines appended to the `.pcof`: the three `tb` groups, a comment line and a truncated `tb O1 H1` line; `niter 2`. `commands.txt`: `epsr 200DMSO400Water.EPSR.inp`, `repeat 3`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Exercised: nearest-angle selection with several angles, `im1 /= im2` neighbour bookkeeping, the purely intramolecular group (no triplet neighbours), the `ia3 /= ia1` rule for different end types, rejection of lines that are not valid `tb` groups, and the run-time output FORMAT for 5 angles (`*(1x,f7.2)`, unlimited repeat). The README quotes a three-body energy of ~712 kJ/mol per molecule in the last `.out`.

Notes

Reproduced: `read_pcof` stores cosines and width 1.0, but `save_files_others` re-reads the `tb` symbol at the end of each iteration and stores angles and width in radians, so later iterations use different triplet energies. The negative amplitude is made positive by `read_pcof` but re-read without `abs`, so it is negative from the second iteration and written back as -0.50.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.